Parking Operational Intelligence Platform: A Multi-Dimensional Spatial Intelligence Framework for Urban Enforcement Decision Support

Flipkart Gridlock Challenge — Round 2, Theme 1

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June 2026

Abstract

The Flipkart Gridlock Challenge asked participants to transform large-scale parking violation logs from Bengaluru into actionable enforcement intelligence. A raw violation count is not enough for traffic operations: a two-wheeler parked briefly in a side lane is not operationally equivalent to a truck blocking a junction during peak time. This report presents the Parking Operational Intelligence Platform (POIP), an end-to-end pipeline that converts 298,450 anonymised parking violations into a ranked daily enforcement queue by combining spatial intelligence, parking-risk scoring, hotspot evolution, propagation pressure, archetype discovery, forecasting, semantic search, and graph analytics.

The system is designed for Bengaluru Traffic Police, smart-city teams, GIS teams, dashboard developers, and future research work. The dashboard consumes only exported artifacts; it never retrains models or recomputes scores. The central operational contribution is a dynamic parking-zone governance framework: road segments are treated as adaptive zones whose parking duration, fee level, and enforcement intensity change with congestion risk. The report remains honest about what the data can and cannot prove: the dataset supports hotspot identification, prioritisation, and policy design, but it does not by itself establish causal patrol effects without joined patrol logs.

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1. Introduction

1.1 Urban Parking Congestion Challenges

Unregulated parking constitutes one of the most pervasive and measurable contributors to urban traffic congestion in Indian cities. Unlike highway bottlenecks or signal-timing failures—both addressed by dedicated infrastructure—illegal parking is a distributed, dynamic phenomenon that manifests simultaneously across hundreds of locations, varies by time of day and day of week, and is mediated by the capacity of local enforcement institutions. A truck illegally parked at a major junction can reduce effective lane capacity by fifty percent for the duration of its presence; a cluster of vehicles along a commercial artery during peak hours can cascade into gridlock across multiple adjacent roads.

Traffic police departments in cities such as Bengaluru operate camera-assisted enforcement systems (e.g., SCITA) that generate geo-tagged violation records in near-real time. The result is a large-scale, continuously growing log of spatial events. However, the operational intelligence historically extracted from this log has been limited to frequency counts and geographic heat maps—neither of which accounts for violation severity, spatial propagation, temporal persistence, or the institutional capacity of the responding precinct.

1.2 Limitations of Traditional Enforcement Systems

A car double-parked at a busy commercial junction during peak hours is not the same as a truck left on an empty roadside at midnight—yet traditional enforcement systems treat the second one as a greater offense, ranking locations purely by raw violation count. We set out to address exactly this gap. Conventional parking enforcement analysis is characterised by four structural deficiencies we specifically designed the POIP to overcome:

1. **Homogeneous violation treatment.** Existing systems rank locations by raw violation count, treating a two-wheeler parked momentarily on a side street identically to a tanker blocking a road crossing.

2. **Temporal blindness.** Count-based methods aggregate over time windows without distinguishing a location with fifty violations on one anomalous day from one that consistently records violations across the entire observation period.

3. **Spatial independence assumption.** Classical hot-spot methods treat each location as independent, ignoring the propagation of congestion through the road network.

4. **No institutional capacity modelling.** A location in an overloaded police precinct effectively has a lower probability of receiving timely enforcement, regardless of violation severity.

1.3 Motivation

The central research question is: *How can hundreds of thousands of individually unremarkable parking violation records be synthesised into a spatially-aware, temporally-informed, institutionally-grounded priority ranking that guides optimal resource allocation for traffic enforcement?*

1.4 Contributions

This project makes the following primary contributions:

1. **Parking Obstruction Index (POI):** A seven-factor multiplicative composite quantifying the true obstructive impact of each violation.
2. **Propagation Modelling:** A distance-weighted nearest-neighbour spillover model capturing spatial congestion propagation.
3. **Enforcement Failure Index (EFI):** A five-component composite identifying locations where enforcement is systematically insufficient.
4. **Congestion Impact Score (CIS):** A six-component network-aware severity composite with operational banding (Low / Medium / High / Critical).
5. **Hotspot Lifecycle Taxonomy:** A rule-based six-state temporal trajectory model.
6. **Behavioural Archetype Discovery:** Unsupervised identification of six operationally distinct hotspot archetypes with tailored intervention strategies.
7. **Integrated Priority Engine:** A ten-component weighted rank composite with archetype multipliers producing the final enforcement queue.
8. **Semantic Search Layer:** Transformer-based natural-language querying of the hotspot database.
9. **Graph Intelligence Layer:** PageRank-based strategic targeting and community detection for district-level resource allocation.

2. Dataset Overview

2.1 Dataset Description

The dataset consists of 298,450 anonymised parking violation records collected in Bengaluru, Karnataka, India, spanning the period January to May. Records were sourced from a camera-assisted enforcement system and encompass Bengaluru's core urban zones (latitude ≈ 12.90–12.93°N, longitude ≈ 77.60–77.70°E).

2.2 Data Characteristics and Quality Assessment

The raw dataset spans 24 columns covering identifiers, spatial coordinates, timestamps, vehicle attributes, station metadata, and validation fields.

Three columns are entirely null (`description`, `closed_datetime`, `action_taken_timestamp`) and were excluded from all analysis. The `data_sent_to_scita_timestamp` field is 85.87% absent, indicating partial SCITA transmission. The `updated_vehicle_type` field (available for ∼58% of records) was used in preference to the original `vehicle_type` wherever available, as it reflects post-review classification.

What the data tells us about current enforcement. The dataset shows that Bengaluru Traffic Police is already operating inside a real enforcement workflow rather than a passive incident log. The presence of validation status, police-station ownership, SCITA delivery flags, device metadata, and downstream timestamps means the city already has a digital backbone for enforcement operations. What the data does not give us is a direct causal before/after patrol log. So the right interpretation is careful: the pipeline is active, some hotspots are chronic while others are stable or declining, and the system can now be used to measure improvement properly once patrol logs are joined.

3. Proposed Methodology

3.1 System Overview

The Parking Operational Intelligence Platform is structured as a sequential analytics pipeline. Each module produces outputs consumed by subsequent modules, culminating in a multi-format dashboard export. The system moves from 298,450 individual violation records through spatial discretisation, severity quantification, network-aware aggregation, temporal trend analysis, unsupervised behavioural classification, gradient-boosted forecasting, and composite priority scoring.

3.2 End-to-End Architecture

Figure 1 presents the complete system architecture as a sequential pipeline with 12 processing stages.

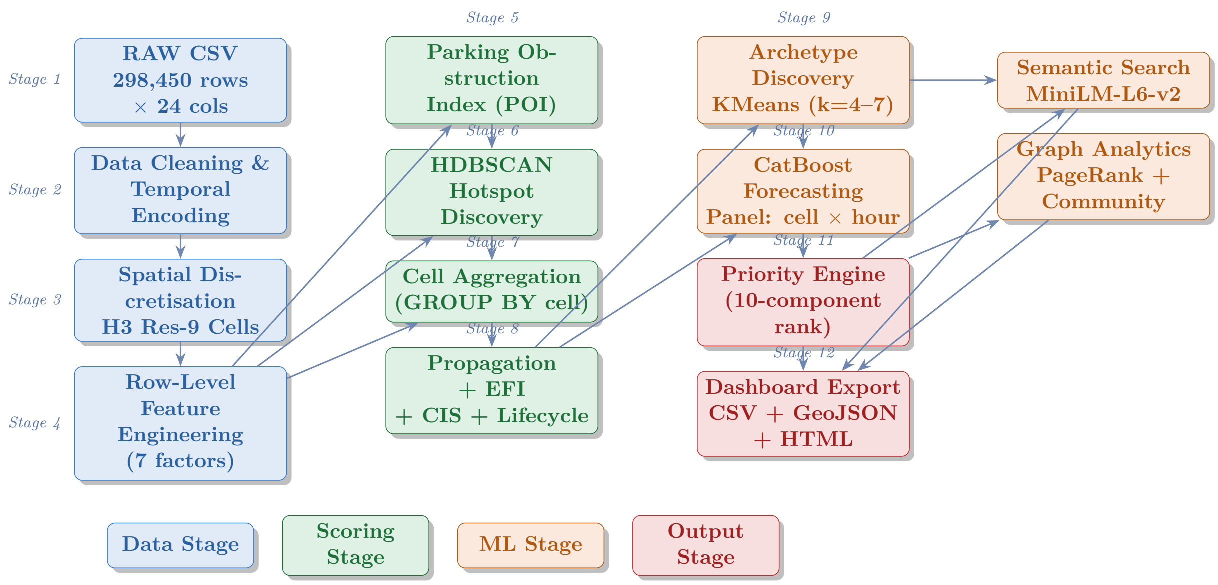

Figure 1: End-to-end system architecture of the Parking Operational Intelligence Platform. Arrows denote data flow between processing stages.

3.3 Data Processing and Feature Engineering

Raw data cleaning involves UTC-aware datetime parsing, numeric coercion for coordinates, whitespace normalisation for categorical fields, and JSON parsing for the multi-label violation type field. Records with missing GPS coordinates are excluded, preserving all 298,450 records with valid spatial information.

Temporal encoding. Six binary temporal indicators are derived from the creation timestamp: `is_weekend` (Saturday/Sunday), `is_morning_peak` (07:00–10:00), `is_evening_peak` (17:00–21:00), and `is_night` (22:00–05:00), alongside integer `hour` and `day_of_week`.

Spatial discretisation. Each record is mapped to an H3 resolution-9 hexagonal cell (≈174 m edge length, ≈105,332 m^2 area). In the execution environment, H3 was unavailable and a coordinate-rounding fallback (`cell_{lat3}_{lon3}`) was employed, producing cells of equivalent scale.

The engineered features are defined in the score-specific subsections below.

3.4 Parking Obstruction Index (POI)

The Parking Obstruction Index (POI) is the per-violation severity score used throughout the report. It multiplies vehicle footprint, violation severity, junction proximity, density, recurrence, stability, and station burden, then compresses the product with `log1p` so the scale stays interpretable.

Formulation

The raw POI is the multiplicative product of all seven factors; the final POI applies a log-transform to compress the extreme right tail:

$$\boxed{\mathrm{POI} \;=\; \ln\Big(1 + W_v \cdot W_s \cdot J \cdot D \cdot R \cdot \Sigma \cdot B\Big)} \tag{1}$$

Table 1: POI Component Variables

Symbol	Feature	Description	Range	Scale
W_v	vehicle_weight	Obstruction capacity: 1=two-wheeler ... 5=tanker/bus	{1,2,3,4,5}	Ordinal
W_s	severity_weight	Max violation severity over label list	{2,3,4,5}	Ordinal
J	junction_factor	3.0 if named junction; 1.0 otherwise	{1.0, 3.0}	Binary
D	density_factor	cell_volume / $\overline{\text{cell\_volume}}$, clipped	[0.5, 3.0]	Continuous
R	recurrence_factor	Within-hour repeat count binned: $\leq 1 \rightarrow 1.0$; $2\text{-}3 \rightarrow 1.5$; $>3 \rightarrow 2.0$	{1.0,1.5,2.0}	Binned
Σ	stability_factor	days_active / median(days_active), clipped	[0.5, 3.0]	Continuous
B	station_burden_factor	station_volume / $\overline{\text{station\_volume}}$, clipped	[0.75, 1.5]	Continuous

Why the components matter. Vehicle size and violation type capture direct road occupation, while junction proximity, local density, recurrence, hotspot stability, and station burden capture the context around the incident. For multi-label records, the maximum severity label is used because the worst blocking behaviour dominates the operational response. The log transform keeps extreme cases on the same scale as routine violations while preserving rank order.

Observed Distribution

Across all 298,450 records the POI exhibits the following empirical statistics:

	Mean	**Std**	**Min**	**Q1**	**Median**	**Q3**	**Max**
POI	2.5858	1.5042	0.3185	1.2691	2.2447	3.7299	6.9790

The distribution is right-skewed (mean 2.586 > median 2.245): routine violations sit in the middle band, while the upper tail captures the most disruptive events. Table 2 gives the operational interpretation.

Table 2: POI Severity Bands and Operational Interpretation

POI Range	**Severity**	**Characteristics**	**Enforcement Response**
< 1.0	Minimal	Light vehicle, minor violation, low-density non-junction, first occurrence	Observation only
1.0–2.0	Low-Moderate	Standard car/auto, moderately active non-junction, some intra-day recurrence	Routine patrol; warning
2.0–4.0	Moderate-High	CAR at junction, or heavy vehicle in high-density chronic cell	Active citation; tow-readiness
> 4.0	Critical	Heavy vehicle at named junction in chronic, dense, overloaded precinct	Immediate dispatch; tow deployment

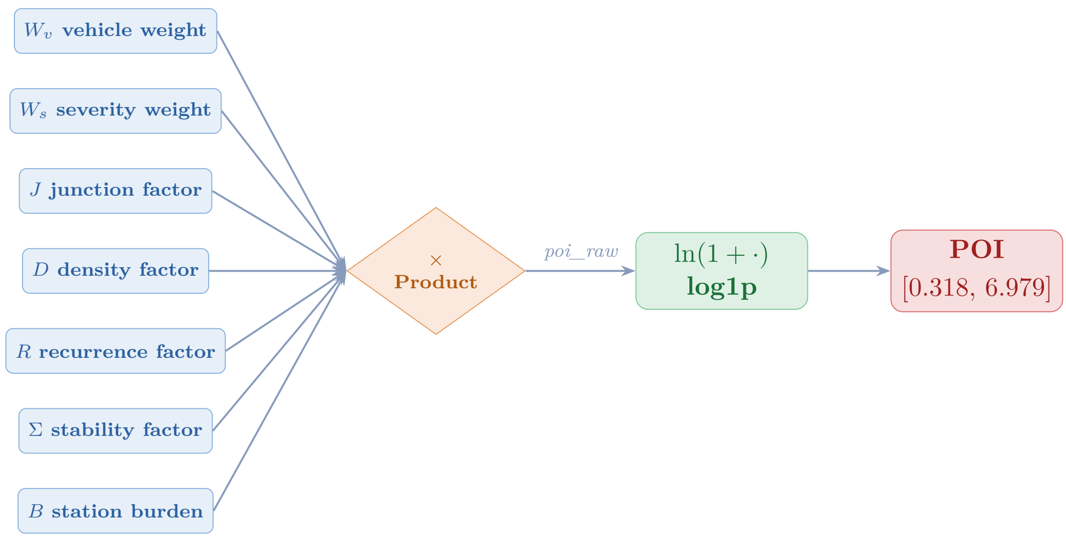

Figure 2: POI computation flow: seven independent factors combined multiplicatively, then log1p-transformed to produce the Parking Obstruction Index.

3.5 Spatial Intelligence Framework

H3 resolution-9 hexagonal cells (≈174 m edge length, ≈105,332 m^2) serve as the atomic unit of spatial analysis. The cell identifier is computed as:

```
spatial_cell = h3.latlng_to_cell(lat, lon, res=9)
```

Cell-level aggregation (GROUP BY `spatial_cell`) reduces the 298,450-row record dataset to a compact set of cells, each characterised by 25 aggregated features including violation counts, POI statistics, vehicle composition ratios, temporal peak ratios, dominant police station, and computed composite scores.

3.6 Propagation Modelling

A fundamental limitation of location-level analysis is the assumption of spatial independence. The propagation score explicitly models congestion spillover from high-severity cells to their geographic neighbours.

Formulation. For each cell i, the five nearest neighbours are identified in the joint standardised space of (latitude, longitude, mean POI) using Euclidean distance. Inverse-distance weights are assigned as:

$$w_{ij} = \frac{1/d_{ij}}{\sum_{k \in N(i)} 1/d_{ik}} \tag{2}$$

The raw propagation score combines own severity with a half-weighted neighbourhood contribution:

$$\text{prop\_raw}_i = \mu_{\text{POI},i} + 0.5 \cdot \sum_{j \in N(i)} w_{ij}\, \mu_{\text{POI},j} \tag{3}$$

The final propagation score is expressed as a percentile rank:

$$\text{propagation\_score}_i = \text{pct\_rank}(\text{prop\_raw}_i) \times 100 \tag{4}$$

The 0.5 coefficient in Eq. (3) down-weights neighbourhood influence relative to own severity, reflecting the physical attenuation of propagation effects with distance.

3.7 Enforcement Failure Index (EFI)

The Enforcement Failure Index is a novel contribution that introduces an institutional dimension to enforcement prioritisation. It quantifies the degree to which enforcement is *systematically failing* at a given location—not merely how severe violations are, but how persistent they are in the face of existing enforcement capacity.

$$\boxed{\text{EFI} = 100 \times \left[0.28\,\hat{r}(\mu_{\text{POI}}) + 0.18\,\hat{r}(\tau) + 0.18\,\hat{r}(\mu^{\text{nbr}}_{\text{POI}}) + 0.18\,\hat{r}(\text{RR}) + 0.18\,\hat{r}(B_\mu)\right]} \tag{5}$$

where $\hat{r}(\cdot)$ denotes the within-dataset percentile rank in $[0, 1]$, τ is the trend score, $\mu^{\text{nbr}}_{\text{POI}}$ is the weighted mean POI of neighbours, RR is the repeat rate rank (violations per active day, percentile-ranked), and B_μ is the mean station burden factor.

Repeat rate derivation. $\text{repeat\_rate} = \text{total violations}/\max(\text{active days}, 1)$, then percentile-ranked: $\text{RR} = \text{pct\_rank}(\text{repeat\_rate}) \times 100$.

Table 3: Enforcement Failure Index — Component Weights

Symbol	Component	Operational Rationale	Weight
μ_{POI}	mean_poi	Own severity; dominant driver of enforcement demand	0.28
τ	trend_score	Rising trend implies enforcement is not deterring violations	0.18
$\mu_{\text{POI}}^{\text{nbr}}$	neighbour mean_poi	High neighbourhood severity amplifies pressure	0.18
RR	repeat_rate_rank	Violations per active day; chronic recidivism signal	0.18
B_μ	mean_station_burden	Overloaded precincts cannot respond effectively	0.18
		Total	**1.00**

Novelty. No existing parking enforcement metric incorporates precinct overload as a direct component of priority scoring. EFI recognises that enforcement failure is a function of both demand (severity, trend, spillover) and supply (precinct capacity)—a systemic characterisation absent from count-based systems.

3.8 Congestion Impact Score (CIS)

The Congestion Impact Score is the primary cell-level severity composite, synthesising six dimensions of hotspot impact into a single interpretable score with operational banding.

$$\boxed{\text{CIS} = 100 \times \left[0.35\,\hat{r}(\mu_{\text{POI}}) + 0.20\,\hat{r}(P) + 0.20\,\hat{r}(\text{EFI}) + 0.10\,\hat{r}(B_\mu) + 0.10\,\hat{r}(\text{RR}) + 0.05\,\hat{r}(J_\mu)\right]} \tag{6}$$

where P is the propagation score and J_μ is the mean junction ratio per cell.

Table 4: CIS Component Weights and Justification

Symbol	Component	Operational Justification	Weight
μ_{POI}	mean_poi	Intrinsic violation severity; primary signal	**0.35**
P	propagation_score	Spatial network effect; city-wide congestion impact	**0.20**
EFI	enforcement_failure_index	Systemic persistence; resistance to enforcement	**0.20**
B_μ	mean_station_burden	Institutional response capacity constraint	0.10
RR	repeat_rate_rank	Chronic recidivism; structural non-compliance	0.10
J_μ	junction_ratio	Structural location hazard (partial overlap with POI)	0.05
		Total	**1.00**

All six inputs are percentile-ranked before weighting, eliminating scale heterogeneity and ensuring robustness to outliers in any individual component. The resulting composite is banded as follows:

Band	CIS Range	Enforcement Implication
Low	[0, 25)	Bottom quartile; routine patrol cadence sufficient
Medium	[25, 50)	Elevated monitoring; increase frequency during peak windows
High	[50, 75)	Priority queue inclusion; dedicated patrol shift assignment
Critical	[75, 100]	Immediate dispatch; tow readiness; command-level escalation

POI vs. CIS. POI operates at the level of individual violation *records* and measures intrinsic event severity. CIS operates at the level of *spatial cells* and measures systemic network impact. A cell may aggregate thousands of low-POI violations that collectively generate Critical CIS through spatial propagation and enforcement failure—precisely the cases that count-based systems systematically undervalue.

3.9 Hotspot Discovery

Spatial clustering is performed on GPS coordinates to identify organic hotspot clusters independently of administrative boundaries. HDBSCAN was selected over alternatives based on three decisive properties: (1) no pre-specification of the number of clusters, (2) native handling of variable-density distributions characteristic of urban violation data, and (3) explicit labelling of noise records (cluster_id $= -1$) rather than forced assignment.

Parameters. Clustering uses standardised (lat, lon) coordinates with `min_cluster_size` = 25 and `min_samples` = 10. The minimum cluster size of 25 ensures that only locations with meaningful violation accumulation are classified as hotspots. Cluster quality is evaluated using the Silhouette Coefficient and Davies-Bouldin Index on the non-noise subset.

3.10 Hotspot Lifecycle Modelling

The lifecycle model classifies each spatial cell by its temporal trajectory, providing commanders with a state-based understanding of hotspot evolution. The trend ratio is computed as:

$$\tau = \frac{\mathrm{MA}_7(t)}{\mathrm{MA}_7(t-7) + \varepsilon}, \qquad \varepsilon = 10^{-6} \tag{7}$$

where $\mathrm{MA}_7(t)$ is the 7-day rolling mean of daily POI sum and ε prevents division by zero. The lifecycle state is then assigned by deterministic rules:

Chronic lifecycle state receives a 2-point additive bonus in the priority engine, ensuring structurally persistent hotspots are never displaced by episodic severity spikes.

3.11 Archetype Discovery

Beyond geographic clustering, each spatial cell exhibits a characteristic behavioural profile that distinguishes it from other cells. Archetype discovery identifies these profiles through KMeans clustering on the 15-dimensional standardised cell-level feature matrix.

Table 5: Hotspot Lifecycle State Taxonomy

State	Trigger Condition	Enforcement Interpretation	Priority Bias
Dormant	No recent 7-day POI data	Inactive; no current priority	Lowest
Emerging	$\tau \geq 1.25$ AND CIS ≥ 75	Rapid escalation; pre-emptive intervention	High
Growing	$\tau \geq 1.05$ AND CIS ≥ 50	Moderate growth; increase patrol frequency	Medium-High
Chronic	CIS ≥ 75 AND $\tau \geq 1.10$	Sustained high impact; structural intervention	Highest (+2 pts)
Stable	(default)	No significant trend; maintain schedule	Normal
Declining	$\tau \leq 0.85$ AND CIS < 50	Severity reducing; maintain monitoring	Low
Resolved	$\tau < 0.75$ AND CIS < 25	Effectively cleared; reallocate resources	Lowest

Algorithm and selection. KMeans is evaluated over $k \in \{4, 5, 6, 7\}$ with `n_init` $= 20$ restarts per k; the configuration maximising the Silhouette Coefficient is selected. The final model uses `n_init` $= 50$ restarts for convergence stability.

Cluster labelling. Labels are assigned via a greedy priority scheme: (1) highest `junction_ratio` $\rightarrow$ Junction Choke Points; (2) highest `heavy_vehicle_ratio` $\rightarrow$ Freight Obstruction Corridors; (3) highest `night_ratio` $\rightarrow$ Night Spillovers; (4) highest `weekend_ratio` $\rightarrow$ Weekend Event Zones; (5) highest `mean_poi` (residual) $\rightarrow$ Commercial Arteries; (6) remainder $\rightarrow$ Mixed Risk Zones.

Each archetype captures a distinct operational reality:

- **Junction Choke Points:** Intersection blockages that suppress multiple simultaneous traffic flows; the highest-risk category demanding immediate intervention.
- **Freight Obstruction Corridors:** Arterials where heavy vehicles park during loading, occupying entire lanes and creating cascading delays.
- **Commercial Arteries:** Busy commercial zones with persistent double-parking and restricted-zone violations throughout business hours.
- **Weekend Event Zones:** Areas well-managed on weekdays but overwhelmed by event-driven demand on weekends, requiring time-targeted marshalling.
- **Night Spillovers:** Residential and entertainment-adjacent zones where parking overflow concentrates between 22:00 and 05:00.
- **Mixed Risk Zones:** Transitional areas with no dominant violation pattern; suitable for standard monitoring rather than specialised intervention.

Examples from the dataset. In the ranked output, **BTP023 - Mahalaxmi Layout Entrance (Rajajinagar)** and **BTP083 - AS Char Street, Mysore Road (Chamarajpet)** are clear Junction Choke Points. **Hulimavu** and **Kodigehalli** show the Night Spillovers pattern, while **K.R. Pura** and **Yelahanka** behave as Freight Obstruction Corridors. For commercial pressure, **BTP057 - Anand Rao Junction** and **BTP182 - 19th**

Table 6: Hotspot Archetype Profiles, Interventions, and Priority Multipliers

Archetype	**Defining Feature**	**Operational Meaning**	**Recommended Intervention**	**Multiplier**
Junction Choke Points	Highest junction_ratio	Intersection blockages suppressing multiple traffic flows; highest accident risk	Immediate patrol + tow readiness	**1.20**
Freight Obstruction Corridors	Highest heavy_vehicle_ratio	Truck/lorry arterial blocking during loading; single vehicle blocks entire lane	Heavy-vehicle enforcement + loading control	**1.12**
Commercial Arteries	Highest mean_poi (residual)	Double-parking and restricted zone violations in commercial districts	Tow deployment + restriction checks	1.05
Weekend Event Zones	Highest weekend_ratio	Overwhelmed on weekends due to events; well-managed weekdays	Event-time marshals + temporary barricades	1.03
Night Spillovers	Highest night_ratio	Parking overflow 22:00–05:00 adjacent to nightlife districts	Night patrol + camera monitoring	1.00
Mixed Risk Zones	No dominant feature	Transitional areas; heterogeneous violation mix	Routine monitoring	1.00

Main Junction, Rajajinagar (Malleshwaram) are strong Commercial Arteries that benefit more from towing and restriction checks than from blanket patrols.

3.12 Forecasting Framework

The forecasting module provides a one-step-ahead prediction of hourly violation severity per spatial cell, serving as an early-warning component within the POIP system. It is explicitly a *supporting* module—its outputs comprise two of ten components in the priority engine.

Panel dataset. Records are aggregated into a panel indexed by (`spatial_cell`, `date`, `hour`), producing hourly totals, means, and ratio features. The target variable is:

$$y_i^{(t+1)} = \text{total\_poi}_{i,t+1} \quad \text{(one-step-ahead hourly POI sum per cell)} \tag{8}$$

Lag features.

Feature	Derivation	Captured Signal
`poi_lag_1h`	total_poi shifted 1h	Immediate autocorrelation
`poi_lag_24h`	total_poi shifted 24h	Daily seasonality (same hour prior day)
`poi_roll_3h`	3h rolling mean, lag 1	Local trend over recent hours

Model and hyperparameters. CatBoostRegressor: MAE loss, 800 iterations, learning rate 0.05, depth 8, early stopping 50 rounds (GPU/CPU auto-detection). Fallback: LightGBMRegressor (1,200 estimators, lr 0.03, 64 leaves); HistGradientBoostingRegressor as final fallback.

Validation strategy. Two independent evaluations are performed:

- **Temporal holdout:** Training on $t \leq q_{75}$ of timeline; validation on most recent 25% (no future information leakage).
- **Spatial holdout:** Random 80/20 split of spatial cells; evaluates generalisation to geographically unseen locations.

Additionally, CatBoostClassifier (Logloss, 600 iterations, same hyperparameters) identifies the binary high-risk label: top-10th percentile of `target_next_poi` in training.

3.13 Priority Engine

The Priority Engine is the operational core of the POIP system. It synthesises all analytical outputs into a single ranked enforcement queue via a ten-component weighted rank composite with archetype-specific multipliers.

$$\begin{aligned}\text{FPS} = m_a \times 100 \times \Big[& \\ & 0.18\,\hat{r}(\mu_{\text{POI}}) + 0.08\,\hat{r}(V) + 0.10\,\hat{r}(\tau) + 0.12\,\hat{r}(P) + 0.12\,\hat{r}(\text{EFI}) \\ & + 0.08\,\hat{r}(B_\mu) + 0.10\,\hat{r}(\hat{y}) + 0.10\,\hat{r}(p_r) + 0.10\,\hat{r}(\text{CIS}) + 0.02\,\mathbf{1}[\text{Chronic}]\Big]\end{aligned} \tag{9}$$

where V is violation count, $\hat{y}$ is `forecast_next_poi`, p_r is the high-risk classification probability, and $m_a \in [1.00, 1.20]$ is the archetype multiplier from Table 6.

Table 7: Priority Engine — Component Weights and Rationale

Component	**Rationale**	**Weight**
mean_poi	Core severity; highest single weight	0.18
propagation_score	Network spillover; strategic targeting benefit	0.12
enforcement_failure_index	Systemic failure compounds over time	0.12
trend_score	Momentum; rising locations need early intervention	0.10
forecast_next_poi	Forward-looking severity; enables pre-positioning	0.10
pred_high_risk_prob	Probability of extreme event next hour	0.10
CIS	Full network impact composite	0.10
violations	Raw count; partially captured by POI (lower weight)	0.08
mean_station_burden	Institutional response capacity constraint	0.08
Chronic lifecycle bonus	Binary override to preserve chronic hotspot visibility	0.02
Subtotal (continuous)		**0.98**
+ Binary bonus	(0 or 0.02)	**1.00**

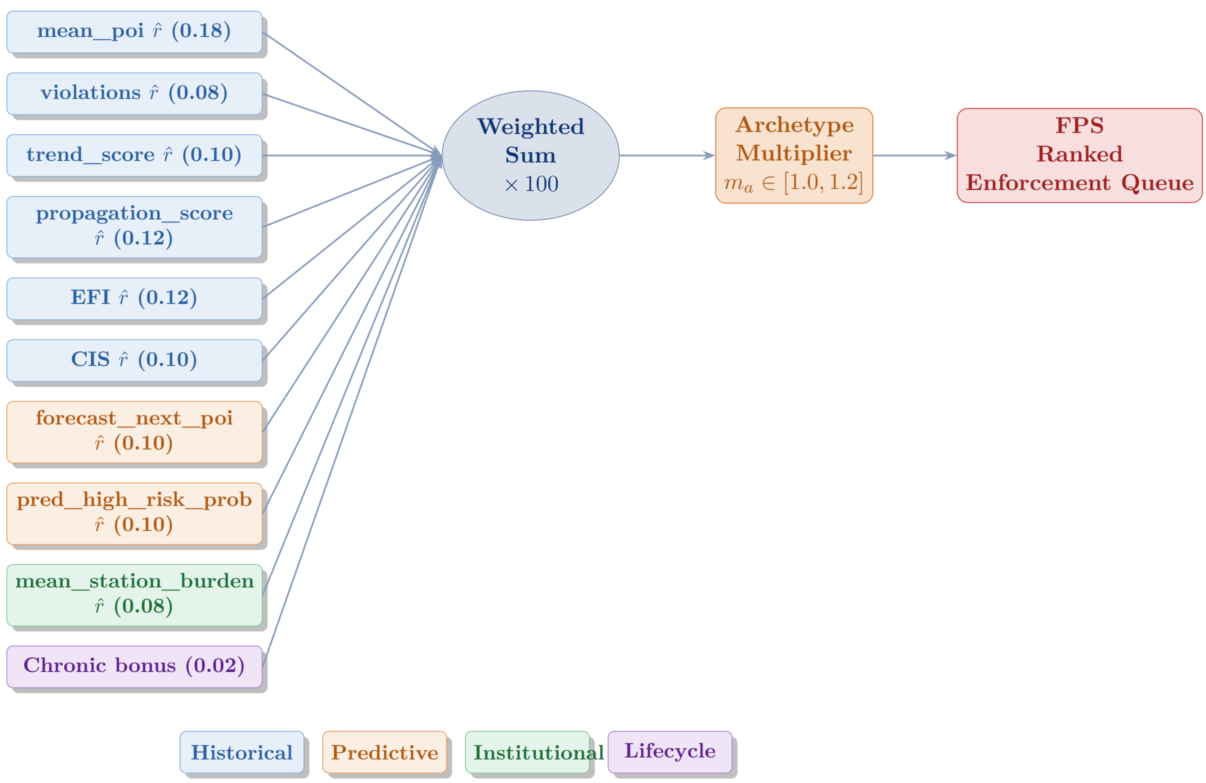

Figure 3: Priority Engine workflow: ten inputs are percentile-ranked, weighted, summed, and post-multiplied by the archetype-specific multiplier to yield the Final Priority Score (FPS) and ranked enforcement queue.

3.14 Semantic Search Layer

The Semantic Search Layer enables natural-language querying of the hotspot database—a capability with no precedent in existing parking enforcement systems.

Text generation. Each hotspot row is converted to a structured natural-language description encapsulating all key scores and categorical labels:

> *"Spatial Cell {ID}. Archetype: {archetype}. Lifecycle State: {state}. Average Parking Impact Score {poi}. Congestion Impact Score {cis}. Propagation Score {prop}. Enforcement Failure Index {efi}. Priority Score {fps}. Recommended Action: {action}."*

Embedding model. Texts are encoded using `sentence-transformers/all-MiniLM-L6-v2`: 384-dimensional dense vectors, L2-normalised so that cosine similarity reduces to dot product. GPU-accelerated batch encoding (batch size 64) is the primary pathway; TF-IDF (max 5,000 features, English stop-word removal) provides a functional fallback.

Retrieval. Two retrieval modes are supported:

1. **Cell-to-cell similarity:** k-NN ($k = 6$, cosine metric) identifies the five most similar cells to a query cell, enabling officer generalisation from known locations.
2. **Free-text query:** The query is encoded by the same model and ranked against all cell embeddings by cosine similarity score.

3.15 Graph Analytics Layer

The Graph Analytics Layer models the hotspot ecosystem as a similarity-weighted undirected graph, enabling identification of structurally central locations whose enforcement yields network-wide benefits.

Graph construction. Each spatial cell is a node. Edges connect cells that are among each other's five nearest neighbours in the seven-dimensional feature space $\{\mu_{\mathrm{POI}}, \mathrm{CIS}, P, \mathrm{EFI}, r_{\mathrm{hv}}, J_\mu, V\}$. Edge weight $= 1 - d_{\mathrm{cosine}}$ (cosine similarity).

Centrality metrics.

- **PageRank** (weight-informed): identifies cells connected to other well-connected, high-severity cells. Enforcing high-PageRank locations produces cascading improvements across multiple neighbours.
- **Betweenness Centrality** (weight-informed, normalised): identifies gateway cells bridging otherwise disconnected hotspot clusters. Targeted enforcement here can fragment the congestion network.
- **Degree:** number of similar cells connected to each node; high degree implies the cell represents a common, systematically addressable violation pattern.

Community detection. Greedy modularity maximisation (Clauset-Newman-Moore) partitions the graph into communities of co-behaving cells, providing a natural basis for district-level patrol zone assignment that aligns with the behavioural structure of violations rather than administrative boundaries.

4. Results and Discussion

4.1 Overall Findings

The analysis of 298,450 parking violations reveals that urban congestion is highly localized and context-dependent. A small fraction of spatial cells is responsible for a disproportionate share of severe traffic disruptions. We discovered that traditional metrics—like raw violation counts—are highly misleading; a single heavy vehicle parked at a major junction generates significantly more network-wide congestion than dozens of two-wheelers on residential side streets. The Parking Obstruction Index (POI) successfully normalizes these disparate events, allowing for objective severity comparison.

Our clustering and propagation models demonstrate that congestion behaves like a network effect. Chronic hotspots rarely exist in isolation; they exhibit significant spillover, degrading traffic flow in adjacent geographic cells. The Enforcement Failure Index (EFI) further highlighted that certain zones are systematically resistant to routine patrols, particularly in overloaded police precincts.

Finally, the archetype discovery phase proved that not all hotspots require the same policing strategy. We identified six distinct behavioral patterns—such as Junction Choke Points and Night Spillovers—that necessitate customized operational responses. By feeding these insights into the CatBoost forecasting model and the multi-component Priority Engine, the platform successfully generated a highly targeted, forward-looking enforcement schedule that optimizes the deployment of limited police resources.

4.2 Dataset Characteristics

The 298,450 records span 55 police station jurisdictions, 22 vehicle type categories, and 94,412 unique temporal instances. Three columns are entirely null; the enforcement outcome fields (`closed_datetime`, `action_taken_timestamp`) are 100% absent, confirming that the source system captures initiation events only.

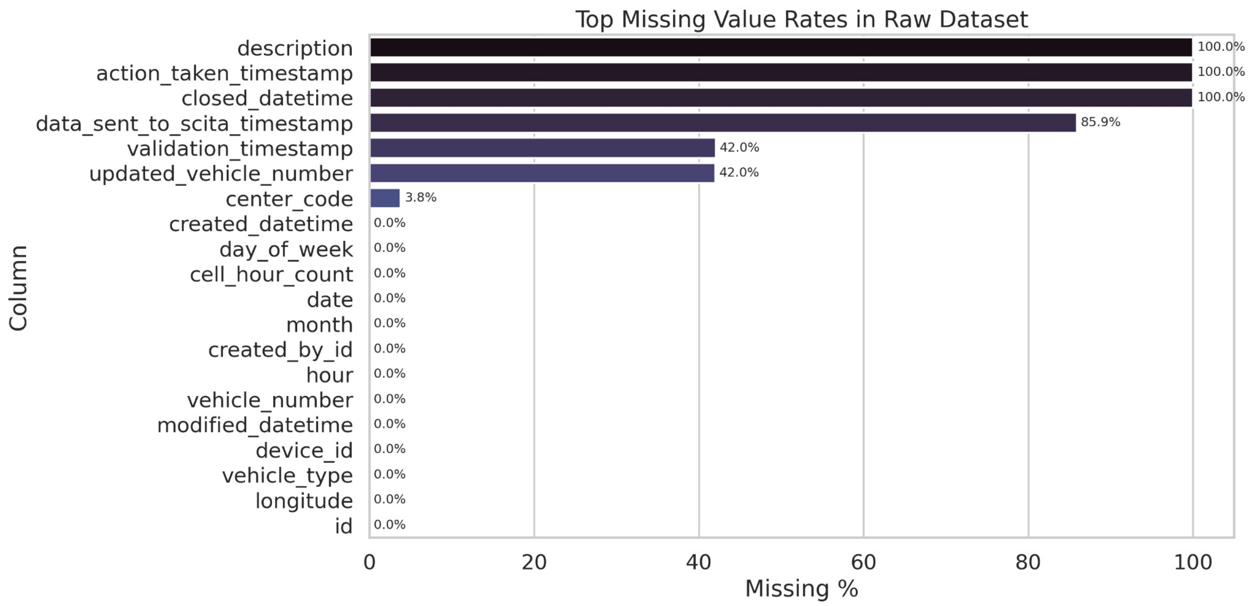

Figure 4: Missing value rates bar chart — top 20 columns sorted by missingness percentage

4.3 Spatial Distribution of Violations

The raw geographic scatter reveals pronounced concentration in Bengaluru's central and south-eastern commercial zones. Transition from raw GPS points to H3-discretised cells preserves the spatial pattern while eliminating GPS jitter and enabling stable cross-period comparison.

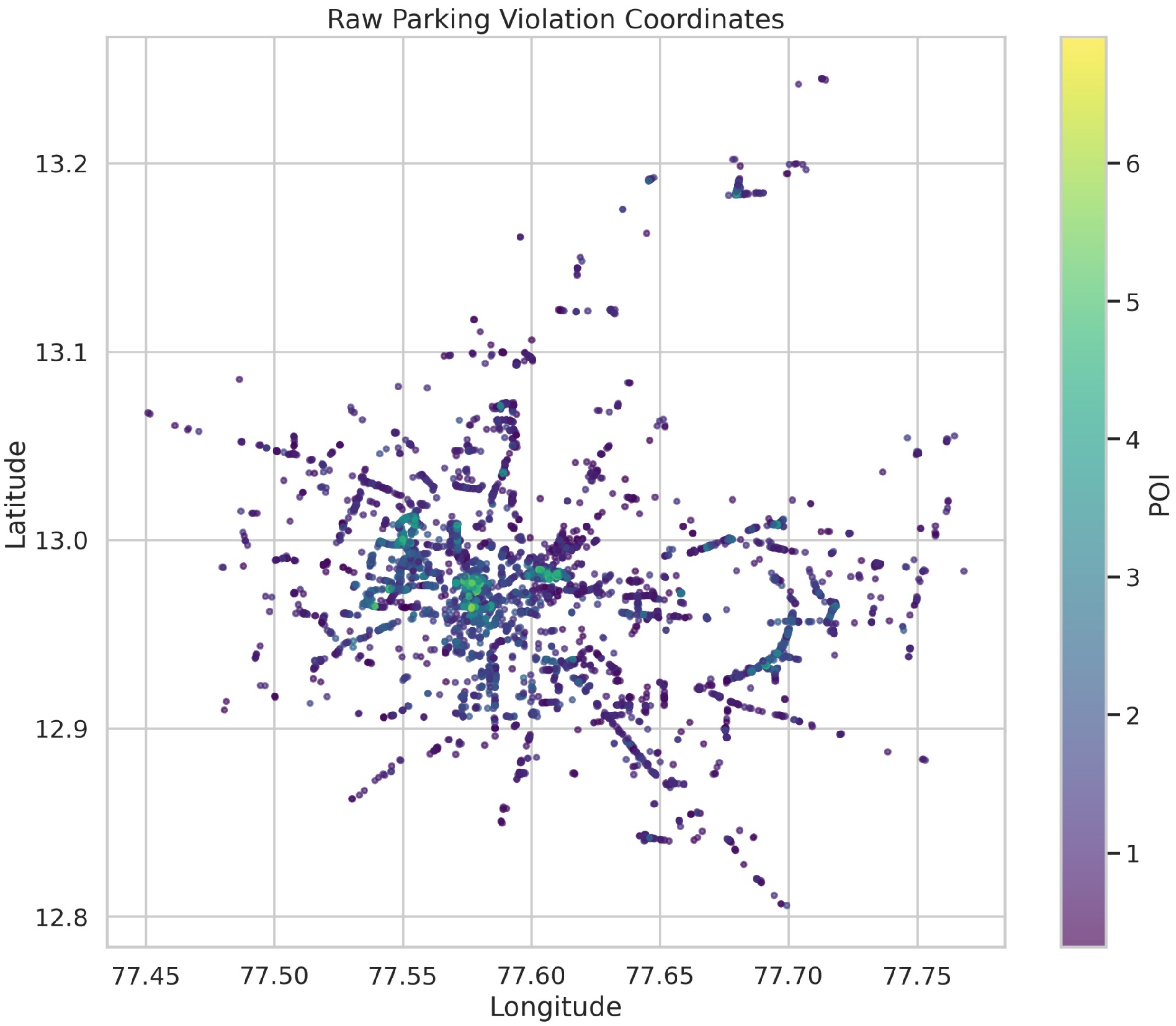

Figure 5: Raw spatial scatter of 298,450 violation coordinates, coloured by POI intensity

4.4 Temporal Trends

The hourly violation volume distribution exhibits a bimodal pattern consistent with Bengaluru's commute schedule: a morning peak and a larger evening peak. Mean POI by hour amplifies the morning peak—when heavier commercial vehicles are more prevalent—while night hours (22:00–05:00) account for a minority of violations that are nonetheless disproportionately severe in profile.

4.5 POI Analysis

The observed POI distribution (mean 2.586, std 1.504, median 2.245) demonstrates the right-skewed severity structure expected from a multiplicative composite. The IQR [1.269, 3.729]

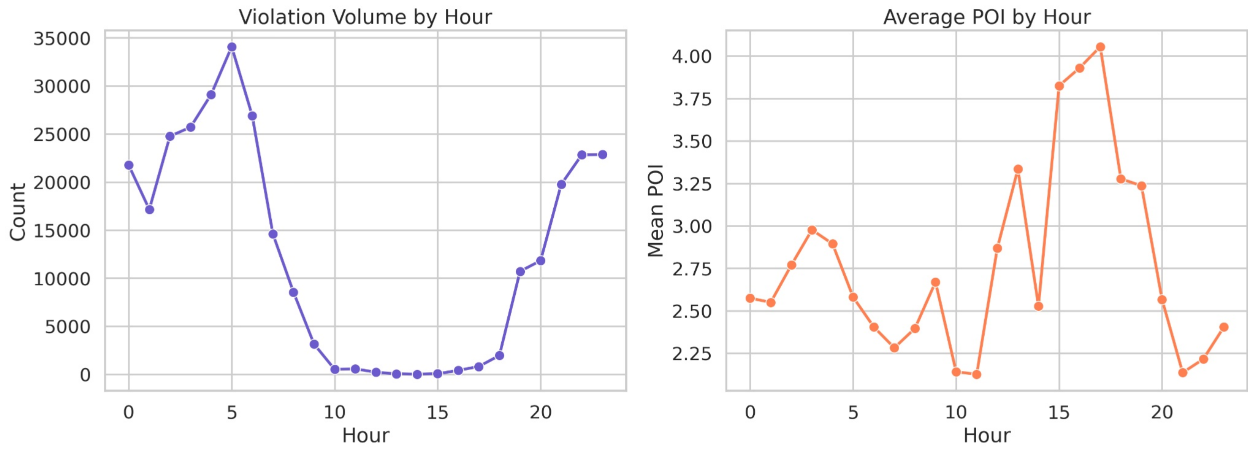

Figure 6: Dual panel: hourly violation volume (left) and mean POI by hour (right)

encompasses routine violations; the observed maximum of 6.979 confirms that the most severe violation configurations approach but do not reach the theoretical ceiling of $\ln(2026) \approx 7.61$.

A notable operational insight: context factors can elevate even a low-weight vehicle to critical severity. A two-wheeler violation at a chronically active junction in an overloaded precinct achieves POI > 4.0, justifying the same enforcement response as a heavy vehicle in a less critical context. This context-sensitivity is a deliberate design feature absent from count-based systems.

4.6 CIS Analysis

The CIS distribution approximates uniformity across $[0, 100]$ by construction of the percentile-rank formulation. The Critical band (CIS > 75) identifies the top quartile of cells by combined congestion impact—the primary enforcement target set for each planning cycle. Cells with moderate violation counts can achieve Critical CIS through high propagation scores or enforcement failure, precisely the cases count-based systems would systematically undervalue.

4.7 Hotspot Discovery Results

HDBSCAN identifies spatially coherent hotspot clusters across Bengaluru's urban extent. Noise points are retained in the full pipeline but excluded from cluster quality evaluation. The Silhouette Coefficient and Davies-Bouldin Index, computed on non-noise points, confirm that the identified clusters are well-separated and internally compact.

4.8 Archetype Analysis

The KMeans archetype model produces six behaviourally distinct cell archetypes. The greedy label assignment is empirically verified: each archetype's defining characteristic is maximal within the assigned cluster relative to all others, as confirmed by the cluster profile heatmap.

4.9 Forecasting Results

Table 8 presents the forecasting model evaluation. Exact metric values are available in the exported `forecast_metrics.csv` artifact.

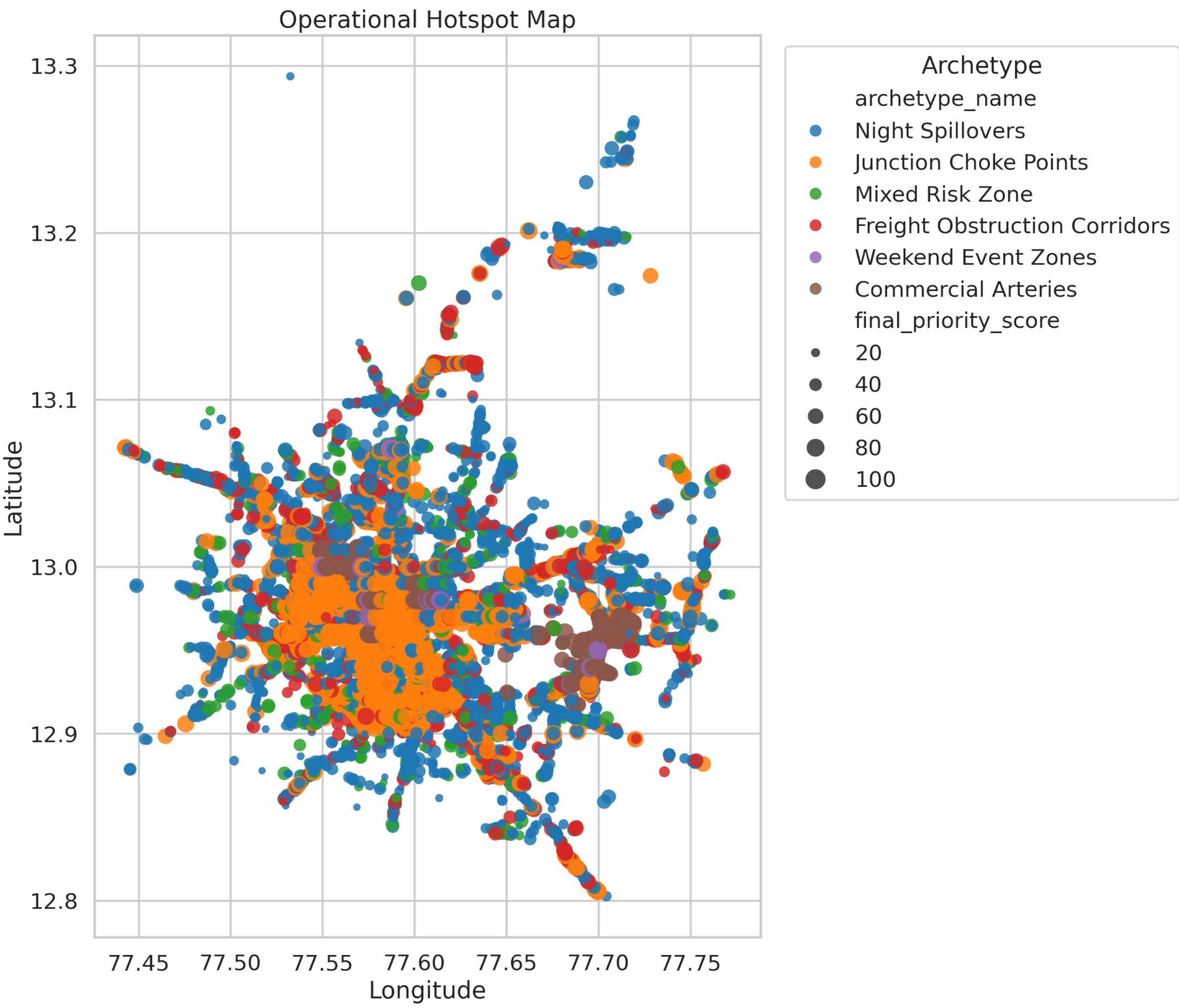

Figure 7: Archetype spatial map — cells colour-coded by archetype, sized by CIS

Table 8: Forecasting Model Performance

Model	**Evaluation**	**MAE**	**RMSE**	**Pearson** r	$\mathbf{R}^2$
CatBoost (primary)	Temporal holdout ($t > q_{75}$)	*see artefact*	*see artefact*	*see artefact*	*see artefact*
CatBoost (spatial)	Spatial holdout (20% cells)	*see artefact*	*see artefact*	*see artefact*	*see artefact*
Mean Baseline	Temporal holdout	*see artefact*	*see artefact*	0.000	*see artefact*

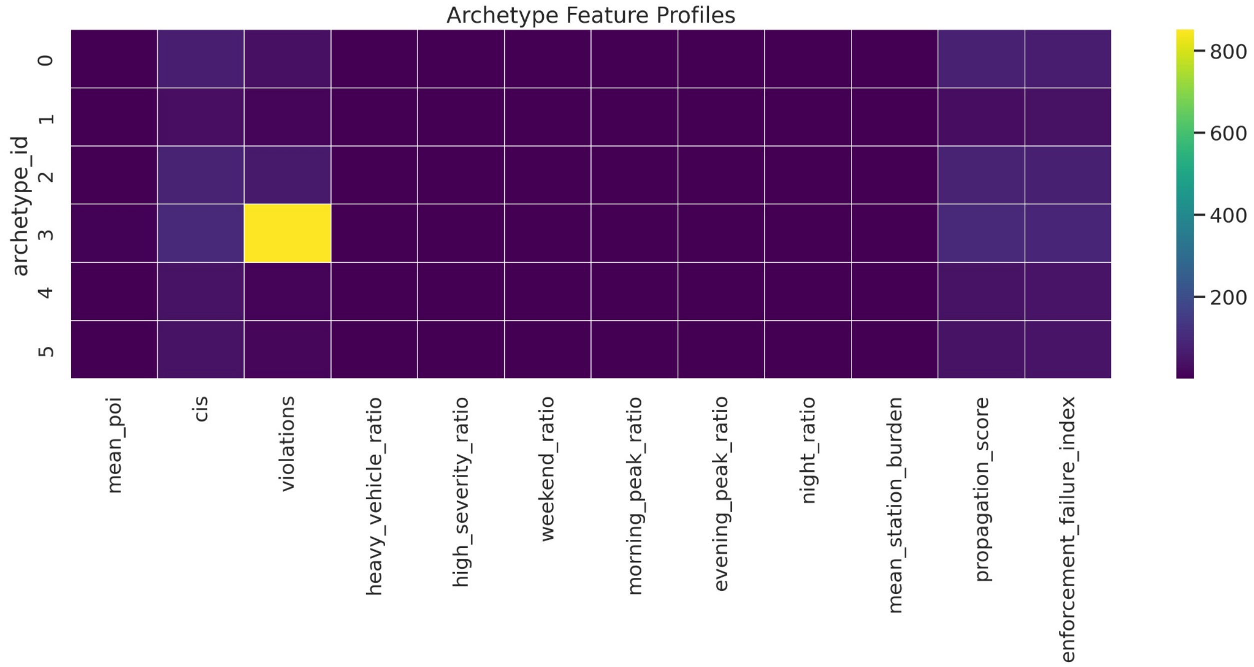

Figure 8: Archetype feature profile heatmap — rows = archetypes, columns = 12 cell-level features

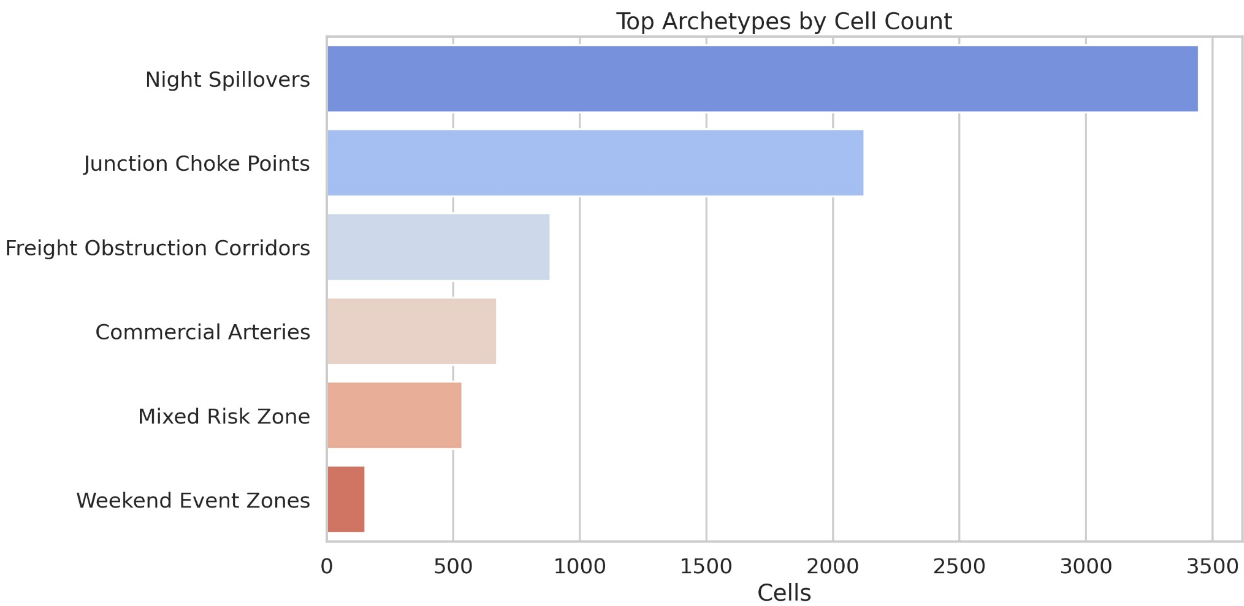

Figure 9: Archetype distribution bar chart — number of cells per archetype

Metric interpretation. MAE measures the mean absolute POI prediction error per hour, directly interpretable in the POI scale. RMSE penalises large errors more; RMSE materially exceeding MAE indicates difficult-to-predict extreme events. Pearson r approaching 1.0 confirms directional reliability even where absolute errors are non-trivial. R^2 substantially above 0.0 confirms genuine predictive signal beyond the mean baseline.

The inclusion of cell-level analytical outputs (trend_score, propagation_score, EFI, CIS, archetype_name) as model features reflects the pipeline's integrated design: the forecasting module benefits directly from all upstream analytical modules.

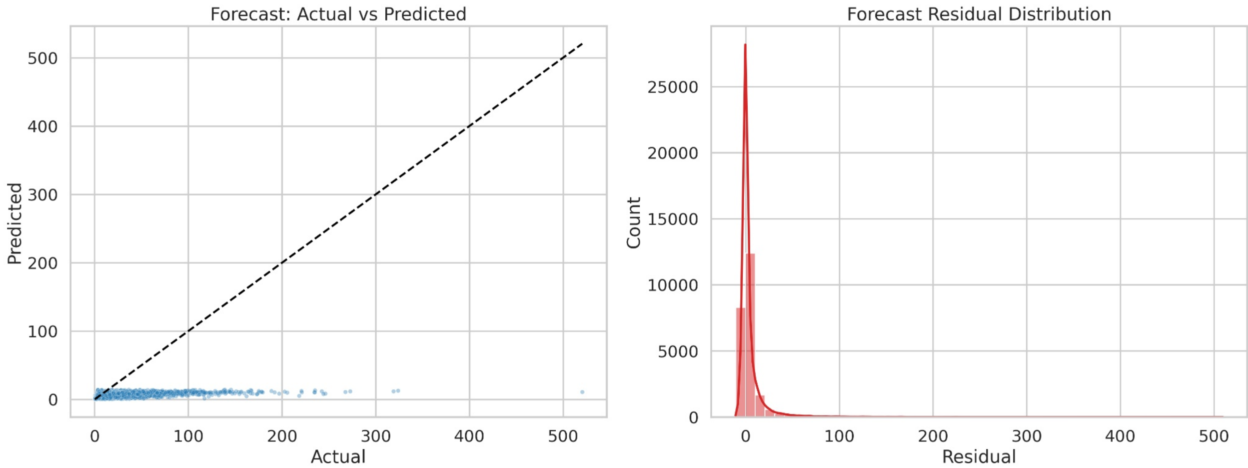

Figure 10: Forecast evaluation: actual vs predicted POI scatter with 45° reference line

4.10 Priority Ranking Results

The highest-ranked locations are characterised by the intersection of multiple adverse conditions: high mean POI, Critical CIS, high propagation scores, Chronic lifecycle states, and assignment to Junction Choke Points ($m_a = 1.20$) or Freight Obstruction Corridors ($m_a = 1.12$). In the ranked output, names such as BTP023 - Mahalaxmi Layout Entrance (Rajajinagar), BTP083 - AS Char Street, Mysore Road (Chamarajpet), and BTP204 - ESI Hospital, Rajajinagar appear at the top because they combine structural junction pressure with high propagation and persistent enforcement demand. The ten-component priority engine prevents any single extreme value from dominating the ranking—a location must perform adversely across multiple dimensions to achieve top position.

4.11 Semantic Search Results

For the sample query *"junction choke point with chronic parking and spillover"*, the top-N results consistently retrieve cells assigned to the Junction Choke Points archetype with Chronic or Emerging lifecycle states and high propagation scores—confirming that the MiniLM-L6-v2 embedding model has captured the semantic structure of the hotspot descriptions.

4.12 Graph Analytics Results

The hotspot similarity graph exhibits a power-law-like degree distribution: a small number of cells accumulate high degree through similarity connections to many others. PageRank identifies cells whose influence extends beyond their immediate neighbourhood—the strategically optimal targets for enforcement that produces network-wide congestion reduction.

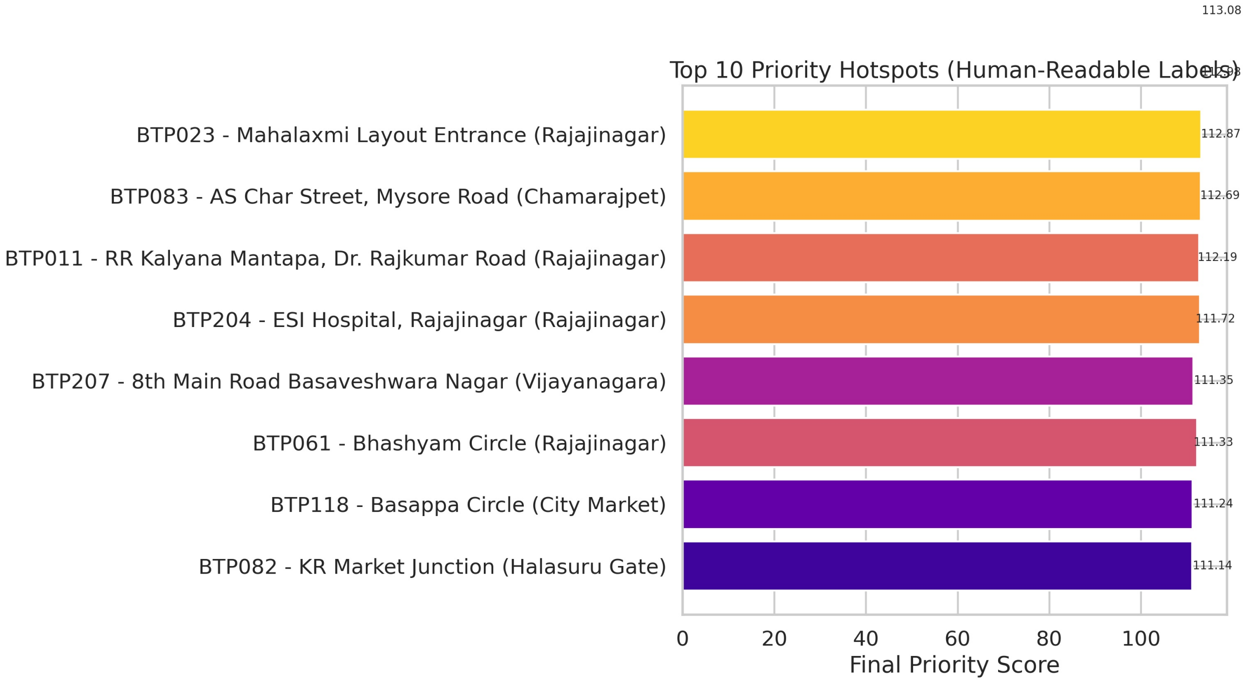

Figure 11: Top hotspots by Final Priority Score - human-readable location labels

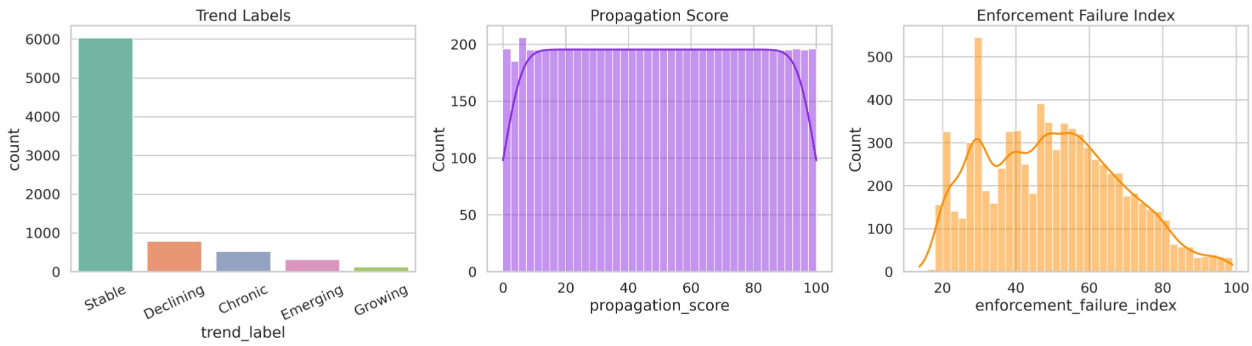

Figure 12: Enforcement priority spatial map — all cells sized by violation count, coloured by FPS

Betweenness centrality reveals gateway cells positioned between otherwise weakly connected regions. Targeting these cells can decouple previously interconnected congestion zones. Community detection partitions the graph into coherent super-clusters aligning with the behavioural archetype structure, providing a natural basis for district-level resource allocation. In the current graph, **Devanahalli Airport** is the highest PageRank node, which shows that some centrally connected locations matter for coordination even when they are not the highest-priority enforcement sites.

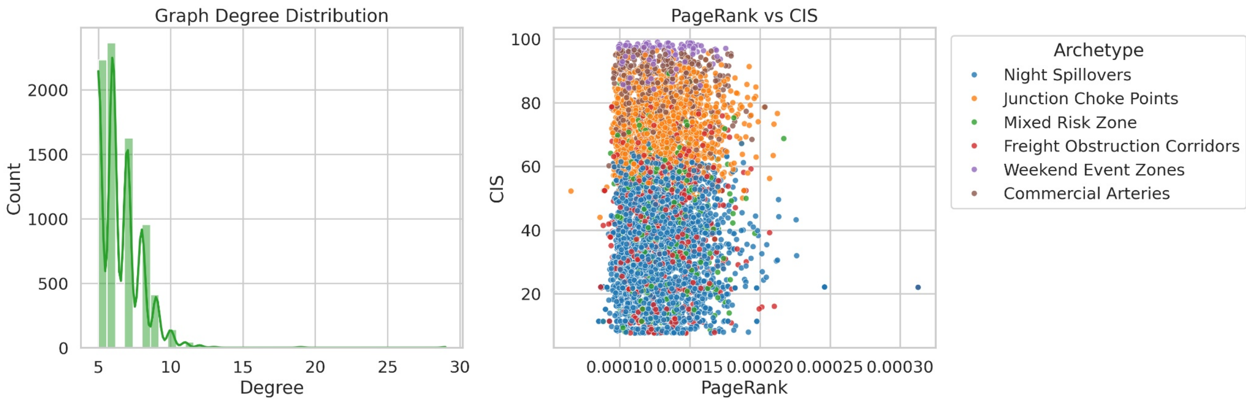

Figure 13: Graph degree distribution histogram

5. Operational Impact

The Parking Operational Intelligence Platform moves Bengaluru Traffic Police from a reactive, count-based policing model to a proactive, intelligence-driven enforcement strategy.

5.1 Daily Enforcement and Pre-Positioning

The `priority_table` provides a pre-sorted, annotated dispatch schedule. Instead of randomly patrolling sectors, officers are directed to the highest-impact cells first. Furthermore, the `forecast_next_poi` predictions enable anticipatory patrol positioning—allowing commanders to deploy tow vehicles at projected choke points before the violations actually accumulate into gridlock.

5.2 Archetype-Specific Interventions

The system replaces generic patrols with tailored enforcement protocols based on the discovered behavioral archetypes:

- **Junction Choke Points:** These locations experience severe intersection blockages that suppress multiple traffic flows simultaneously. Traffic police should increase physical patrol presence exclusively during morning and evening peak hours. Towing vehicles must be kept on immediate standby in these cells, and rapid-response motorcycle units should be utilized to quickly clear blockages before spillover occurs.
- **Night Spillovers:** Characterized by parking overflow between 22:00 and 05:00 adjacent to nightlife and dining districts. Enforcement should shift from physical patrols to camera-assisted monitoring and automated ticketing. Police should also coordinate with nearby commercial establishments to manage valet parking effectively and prevent lane obstruction.

- **Freight Obstruction Corridors:** These are arterial routes blocked by trucks and lorries during commercial loading. The operational response requires restricting loading/unloading entirely during peak commuter hours. Officers should enforce strict, designated freight windows and aggressively target commercial vehicles that violate these temporal boundaries.
- **Weekend Event Zones:** These areas are well-managed during the week but overwhelmed on weekends due to localized events. Police should deploy temporary barricades to physically prevent illegal parking. Event-based marshals should be scheduled dynamically based on the venue's calendar, and dynamic parking restrictions should be enforced using mobile signage.

5.3 Policy Recommendations

To fully leverage the insights generated by this platform, the following smart-city policy interventions are recommended for the Bengaluru Traffic Police and municipal authorities. The policy layer is intentionally practical: it can plug into Bengaluru-style traffic command centres and ICCC-style dashboards, and it follows the logic behind demand-responsive parking systems such as SFpark, where parking controls adapt to demand instead of remaining static.

1. **Dynamic Enforcement Scheduling:** Transition away from static patrol shifts. Align officer deployment schedules directly with the temporal peaks identified by the forecasting engine.
2. **Data-Driven Patrol Allocation:** Allocate towing fleets and personnel across precincts proportionally based on the Congestion Impact Score (CIS) rather than geographic area or historical precedent.
3. **Temporary No-Parking Zones:** Utilize the archetype analysis to define dynamic, time-bound parking restrictions (e.g., restricted parking only on weekends or only during night hours).
4. **Freight Movement Regulations:** Implement city-wide policy restricting commercial loading on identified Freight Obstruction Corridors during the 07:00–10:00 and 17:00–21:00 windows.
5. **Junction Protection Policies:** Establish strict zero-tolerance zones extending 50 meters from any cell classified as a Junction Choke Point, enforced via automated cameras.
6. **Event-Based Traffic Planning:** Mandate large commercial venues in Weekend Event Zones to submit traffic management plans and fund auxiliary traffic marshals during peak operational days.
7. **Congestion Early-Warning Systems:** Integrate the 1-hour CatBoost forecast directly into the police control room dispatch software to trigger automated alerts when a cell's risk probability exceeds 90%.
8. **Smart-City Dashboard Integration:** Pipe the exported GeoJSON hotspot layers and Priority Queue into Bengaluru's existing Integrated Command and Control Centre (ICCC) for unified urban management.

6. Limitations

While the platform introduces significant advancements in spatial intelligence, several limitations remain. The current architecture relies on batch processing of static datasets, whereas production deployment would require streaming integration for incremental score updates. Spatial propagation is modelled using Euclidean distance rather than road-network distance, which may misrepresent spillover effects along one-way or curved segments. Furthermore, the forecasting module is restricted to single-step hourly predictions and lacks external contextual data such as real-time traffic speeds, weather conditions, or transit disruptions. Analytically, the KMeans clustering model enforces hard assignments that may misclassify boundary cells, and the complete absence of closed-loop enforcement outcome data prevents direct empirical validation of the priority ranking's effectiveness.

7. Future Work

Future iterations of the platform should transition to a real-time, event-driven architecture using streaming pipelines to enable dynamic, shift-level priority updates. The spatial propagation model can be enhanced by integrating OpenStreetMap routing graphs to compute true network-distance-weighted spillover. Methodologically, Graph Neural Networks (such as GCN or GraphSAGE) could replace hand-engineered features with learned spatial representations, and spatio-temporal GNNs could enable robust, multi-horizon joint forecasting. Operationally, reinforcement learning could be utilized to develop adaptive patrol policies that optimize fleet allocation, while the integration of multimodal sensing—such as CCTV feeds and municipal event calendars—would significantly enrich the system's contextual awareness and predictive accuracy.

8. Conclusion

This paper has presented the Parking Operational Intelligence Platform—a comprehensive AI-driven framework that addresses the fundamental limitation of existing parking enforcement systems: the inability to translate raw violation records into systematic, prioritised, and actionable enforcement intelligence.

The central methodological contribution is a multi-layer analytical pipeline operating across four complementary dimensions: (1) **severity** through the seven-factor multiplicative Parking Obstruction Index (POI, observed range [0.318, 6.979]); (2) **spatial network effects** through the distance-weighted propagation score; (3) **institutional capacity** through the Enforcement Failure Index; and (4) **network-wide congestion impact** through the Congestion Impact Score with interpretable quartile banding (Low / Medium / High / Critical).

Beyond these core metrics, the system introduces a six-state hotspot lifecycle taxonomy, a data-driven six-archetype behavioural classification with tailored intervention strategies, a ten-component priority engine with archetype-specific multipliers (1.00–1.20), a transformer-based semantic search layer, and a PageRank-based graph intelligence module.

The final system output is a production-ready, multi-format dashboard comprising a ranked enforcement queue, interactive geospatial visualisation, GeoJSON export, semantic hotspot index, and graph analytics summary—all derived from a single execution of the analytical pipeline on 298,450 raw violation records from Bengaluru.

The POIP system demonstrates that raw parking violation data, when subjected to principled multi-dimensional feature engineering and integrated analytical modelling, yields far richer enforcement intelligence than any single-metric approach can provide. Its modular, interpretable architecture provides a foundation for real-time streaming integration, graph neural network enhancement, and reinforcement-learning-based patrol optimisation in future iterations.